systems that has been very influential and is now incorporated into NIST's recommendations for ABAC (https://nvlpubs.nist.gov/nistpubs/specialpublications/NIST.SP.800-162.pdf).

> **DEFINITION *XACML*** is the eXtensible Access-Control Markup Language, a standard produced by the OASIS standards body. XACML defines a rich XML-based policy language and a reference architecture for distributed policy enforcement.

The core components of the XACML reference architecture are shown in figure 8.5, and consist of the following functional components:

- A *Policy Enforcement Point* (PEP) acts like a policy agent to intercept requests to an API and reject any requests that are denied by policy.
- The PEP talks to a *Policy Decision Point* (PDP) to determine if a request should be allowed. The PDP contains a policy engine like those you've seen already in this chapter.
- A *Policy Information Point* (PIP) is responsible for retrieving and caching values of relevant attributes from different data sources. These might be local databases or remote services such as an OIDC UserInfo endpoint (see chapter 7).
- A *Policy Administration Point* (PAP) provides an interface for administrators to define and manage policies.